MUDDASIR ALI

GIS ANALYST | ZONING & PARCEL DATA ANALYST| REMOTE SENSING SPECIALIST

Peshawar, Pakistan | Open to Remote Opportunities | LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

I'm a GIS Analyst with a strong focus on zoning and parcel data production, bringing hands-on experience supporting international clients across the USA and Canada. I enjoy working at the intersection of remote sensing and spatial analysis — from satellite imagery classification to vegetation and land-cover monitoring — and have contributed to disaster response and agricultural mapping projects where accurate, timely data genuinely makes a difference.

TECHNICAL SKILLS

GIS Software	ArcGIS Pro, ArcMap, QGIS, Google Earth Engine
Geospatial Analysis	Zoning & parcel analysis, spatial QA/QC, topology validation, digitization, georeferencing, hazard & flood mapping etc
Remote Sensing	Sentinel-2 multispectral imagery, supervised classification (Random Forest), NDVI/NDBI/NDWI, time-series analysis, image stacking, GeoAI
Database & Data	PostGIS (QGIS connection, spatial queries, intersections, table edits), REST Feature Services, GeoJSON, GeoPackage, GPS/KoboToolbox field data, KML/KMZ
Programming	Python (Basic), workflow automation, AI-assisted development/Coding, web scraping
Web GIS	Leaflet, Mapbox, HTML, CSS, JavaScript (basics)
QA Skills	CRS correction, geometry repair, zone-change detection, sliver/overlap/gap detection, attribute & missing-field validation
Soft Skills	Cross-team collaboration, QA auditing, production workflow documentation

PROFESSIONAL EXPERIENCE

CURRENT ROLE — INTERNATIONAL GIS PRODUCTION

GIS Analyst — Nook Office Solutions

September 2025 – Present

- Digitize, update, and QA parcel and zoning shapefiles across international regions including USA and Canada, maintaining 95–100% spatial accuracy.
- Digitized and QA'd 2,000+ zoning/parcel maps to date; recognized as Employee of the Month three times for accuracy and output.
- Research and scrape authoritative municipal/county GIS portals and REST Feature Services to source current zoning and parcel data; identify and log refreshed source links for 7000+ cities while sustaining a 92% KPI.
- Georeference source maps and imagery with RMS-error control and maintain production QA workflows for topology errors, slivers, gaps, overlaps, and geometry issues.
- Developed lightweight geospatial automation tools to reduce repetitive production QA, including CRS correction, geometry validation, zone-change detection, topology-related checks, and final attribute/spelling/gap/missing-field checks.
- Built a REST spatial-data downloader that accepts service endpoints and automates retrieval to local GeoJSON or GeoPackage outputs.
- Built a GIS data-extraction/scraping utility to streamline discovery of city-level GIS directory and service information from web-based resourses.
- Worked with PostGIS locally, connecting QGIS to a PostGIS database, uploading shapefiles, running spatial intersections/selections, and managing table inserts, updates, and deletes.

- Use raster/TIFF reference data to support zoning symbology checks and verify that expected geometries are represented; collaborate with GIS/data-science teams on workflow and data-standard improvements.
- Audit new team members' digitization output for accuracy and consistency as part of internal QA.

FOUNDATIONAL GIS EXPERIENCE (2023–2025) — GOVERNMENT, AGRICULTURE & DISASTER RESPONSE GIS Associate — Excise, Taxation & Narcotics Control Department, KP
March 2025 – September 2025

- Conducted GPS/KoboCollect field surveys to capture property and land data
- Digitized surveyed boundaries and linked spatial data to official records.
- Verified field-collected data against existing records for accuracy and completeness.

GIS Intern — Crop Reporting Services, Directorate of Agriculture, KP *July 2024 – March 2025 |*

- Performed supervised crop classification across 7 merged districts and surrounding areas; digitized crop-land boundaries and prepared final GIS outputs.
- Conducted time-series NDVI analysis and image stacking using Google Earth Engine, ArcGIS Pro, and ArcMap for crop monitoring.
- Compiled and mapped province-wide rainfall and temperature datasets; led AHP/weighted-overlay land suitability analysis for tomato and pea cultivation.

GIS Intern — Provincial Disaster Management Authority (PDMA), KP
May 2023 – June 2024

- Produced flood and landslide hazard maps; digitized flood-affected areas, damaged infrastructure, housing impacts, and reliefwarehouse locations.
- Supported emergency response planning through timely, accurate spatial data delivery during active disaster events.

Earlier GIS Experience: Survey of Pakistan (2023) | Board of Revenue, KP (2022) — exposure to national survey standards, land records, revenue mapping, and government GIS workflows.

SELECTED AUTOMATION & GIS PROJECTS

Geospatial QA Automation Suite: Production-oriented utilities for CRS correction, geometry repair, zone-change detection, sliver identification, overlap detection, and attribute/spelling/gap/missing-field validation.

REST Spatial Data Downloader: Endpoint-driven utility that retrieves spatial services and saves datasets locally as GeoJSON or GeoPackage, supporting repeatable geospatial data acquisition.

GIS Directory / Data Extraction Engine: Web-scraping utility designed to streamline discovery of city-level GIS directory and service information from mapping portals.

FEATURED ACADEMIC PROJECTS

Awaran Earthquake — Pre-Seismic Anomaly Detection: Analyzed 5 geophysical parameters using 3 anomaly-detection models to identify potential pre-seismic signals.

Sindh Floods 2022 — Impact & Damage Assessment: Extracted NDVI, NDBI, and NDWI from Sentinel-2 imagery; applied Random Forest classification with 87% reported accuracy to identify damaged crop zones and produced a flood zonation map.

Peshawar Landfill Site Suitability Study: Applied AHP and weighted-overlay methods to evaluate potential waste-dumping sites in District Peshawar.

Reticulitermes Species Distribution in China: Supported a Chinese postdoctoral researcher by converting documented records of R. aculabialis, R. chinensis, R. flaviceps, and R. labralis into aconsistent spatial dataset and comparative distribution map.

EDUCATION

BS Geomatics (GIS & Remote Sensing) — University of Peshawar | 2020–2024 *Relevant electives: Land Information Systems, Advanced Photogrammetry*

LANGUAGES

English | Urdu | Pashto